

Background

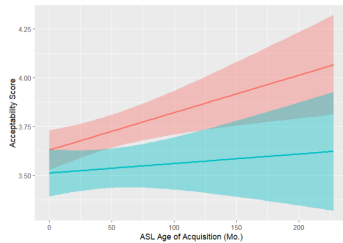
- The word order of ASL, historically debated as either SOV or SVO, reflects linguistic flexibility influenced by English and historical evidence suggesting an original SOV structure.
- Language ideologies within the Deaf community often prefer SOV to emphasize ASL's distinction from English, shaped by social and educational factors.
- This study examines whether Deaf individuals perceive SOV as more representative of ASL through a grammatical judgment task.

Research Questions

- Do Deaf individuals perceive either SVO or SOV ASL sentences as more representative of ASL?
- Do individual differences in Deaf individuals (e.g. age of acquisition and parental hearing status) have an effect?

Data & Analysis

Figure 1

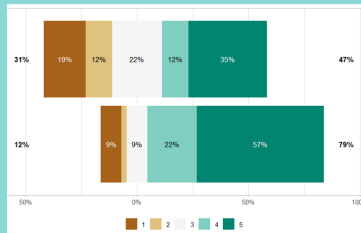


Age of acquisition of ASL resulted in higher acceptability scores for both SOV and SVO sentences.

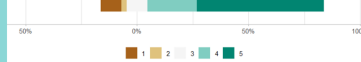
Having deaf parents resulted in lower scores for SVO sentences.

Analysis of Selected Items

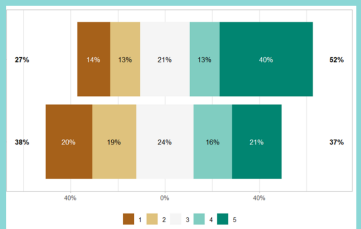
Boy Car Drive
(SOV)



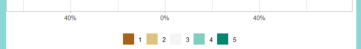
Boy Drive Car
(SVO)



Man Tree Climb
(SOV)



Man Climb Tree
(SVO)



Demographic Effects:

- Age of ASL Acquisition:
 - Older age of acquisition predicted higher acceptability scores for both SOV and SVO sentences (Figure 1).
 - Age of acquisition had a statistically significant effect on acceptability (Effect of Age of Acquisition on Acceptability Score, $p < 0.01$).
- Parental Hearing Status:
 - Participants with Deaf parents rated sentences lower overall compared to those with hearing parents.

SOV sentences were rated more favorably than SVO sentences across the board, but participants with Deaf parents had a slightly higher preference for SVO sentences compared to those with hearing parents.

Participants & Recruitment

Participants:

- 86 Deaf individuals recruited from across the United States.
- Final sample: 82 participants (4 dropped due to incomplete demographics).
- Average age: 34 years (range: 21-55, SD = 5.16).
- Age of ASL acquisition: Mean = 42.84 months (range: 0-228, SD = 64.7).
- 48 participants (just over half) were raised by hearing parents.
- 70 participants reported some degree of college education, with 35 holding a master's degree.

Recruitment:

- Snowball sampling conducted via social media.
- Recruitment flyer outlined criteria:
 - Deaf individuals.
 - Proficient in ASL.
 - Aged 18-40.
- Public platform used to ensure accessibility and target the Deaf community.

Methodology

Materials

Sentence stimuli included both SOV and SVO structures to assess grammatical judgments.

1. Categories Tested:

- Word Order: SVO vs. SOV placement of objects and locations.
- Verb Choice: Plain verbs vs. classifier verbs.
- Prepositions: Analyzed their use in signed sentences.

Procedure

Participants viewed ASL sentences and gave the sentence a rating between 1-5, with 1 being highly unacceptable and 5 being highly acceptable.

Likert scale



Discussion

Preference for Word Order:

Participants showed varied preferences for SOV and SVO sentence structures.

- Example 1: SOV sentence ("BABY COOKIE EAT") received a higher favorable rating (60%) compared to the SVO sentence ("BABY EAT COOKIE," 53%).
- Example 2: SOV sentence ("BOY CAR DRIVE") received 79% favorable ratings, significantly higher than the SVO sentence ("BOY DRIVE CAR"), which showed a more uniform distribution and a higher rejection rate (19%).
- Example 3: SVO sentence ("MAN CLIMB TREE") was preferred (52% favorable ratings) over the SOV version ("MAN TREE CLIMB," 37%), with fewer negative ratings for the SVO form.

Several key themes:

- Efficiency and Predictability** (Monsell & Haggard, 1989; Levy, 2008; Caselli, Hall & Dufour, 2017; Caselli et al, 2016)
- Processing Efficiency** (Brentari, 1998; Padden, 1988; Perlmutter, 1992; Sandler, 2006; Carlson et al., 2016; Hamilton, 2022)
- Predictability and Cognitive load** (Douglas et al)

Participant Preferences for SOV vs. SVO:

- Participants showed a clear preference for SOV structures, perceiving them as more representative of ASL. (Example 1 and 2)

In contrast, MAN CLIMB TREE (SVO) was preferred over MAN TREE CLIMB (SOV), reflecting variability based on sentence context.

Processing Efficiency: Preference for SOV in BABY COOKIE EAT may stem from the semantic relationship between words, aiding predictability and reducing cognitive load.
Phonological Efficiency: SOV structures like BOY CAR DRIVE involve smoother physical transitions between signs, favoring economy of movement. In MAN CLIMB TREE, SVO preference likely reflects efficient transitions between MAN and CLIMB.

Selected citations

1) Brentari, D. (1998). A prosodic model of sign language phonology. Cambridge, MA: MIT press. 2) Fischer, S., & Li, C. 1975. Word order and word order change. Austin: The University of Texas Press. 3) Monsell S., Doyle M. C., Haggard P. N. (1989). Effects of frequency on visual word recognition tasks: Where are they? Journal of Experimental Psychology: General, 118, 43-71. 4) Hill, J. C. 2012. Language attitudes in the American deaf community. Washington, D.C.: Gallaudet University Press.